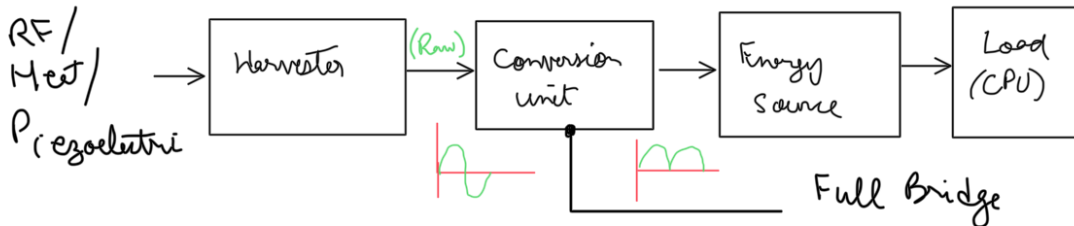
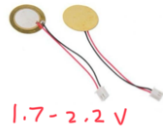


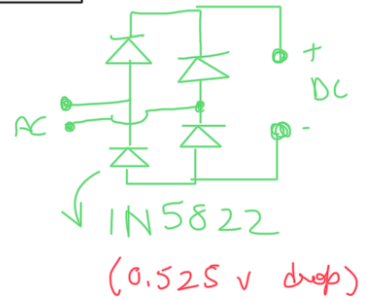
1 Theory Energy harvesting



Piezoelectric Disc →



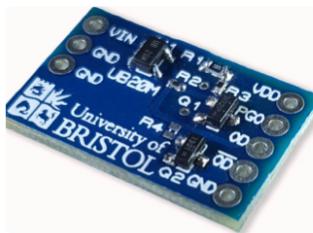
Full Bridge Rectifier



Heat → Peltier Module (1V) each (Not enough)

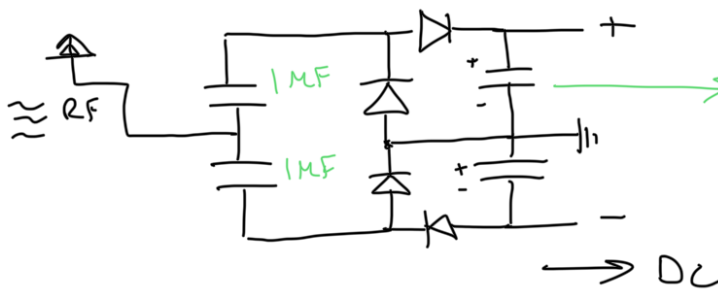
Radio Frequencies → ① High frequency rectifier

VB20M IC powered by the energy of IR



Threshold - 631 - 650 mV

~~IN4001 (1V)~~ BAT54 HY (0.3V) Schottky



50M electrolytic caps

2. Target

Natural

- ① Cosmic & Terrestrial sources = 160 GHz to 30 MHz
- ② Atmospheric Noise = 10 KHz - 10 MHz
- ③ Earth's Magnetic field = 3 KHz - 30 KHz

Man Made

① Communication system

$$\text{AM} = 540 \text{ kHz} - 1.6 \text{ MHz}$$

$$\text{FM} = 88 \text{ MHz} - 108 \text{ MHz}$$

$$\text{TV Broadcast} = 54 \text{ MHz} - 806 \text{ MHz}$$

$$\text{Mobile (3G-5G)} = 700 \text{ MHz} - 6 \text{ GHz}$$

② Microwaves = 2.45 GHz

③ Wi-Fi = 2.4 GHz > 5 GHz > 6 GHz

Most common RF (RF of interest)

★ → AM waves 530 kHz ~ 1.7 MHz

For long range - Medium Wave signal is most reliable

★ FM = 88 - 108 MHz

Power that can be collected → $P_R = P_T G_T \left(G_R \frac{1}{4\pi d} \right)^2$

Power Received Power of transmitter Gain of transmitter Gain of receiving antenna (Unknown)

~50W High

Harvester →

① Impedance Matching

→ Maximum power transfer

Mobile Networks → 900 - 2.5 GHz

+ Wi-Fi → 2.4 GHz - 5 GHz

Final Target Frequency →

900 - 3.5 GHz

Antenna → Monopole Antenna

4-8 cm

$$Z_0 = 60 \ln \left(\frac{4h}{d} \right)$$

↓
50 Ω

h → height of antenna
d → diameter of conductor
Z₀ → impedance of antenna

3. Calculations

Needed -
single Cycle $\rightarrow$

CPU $\rightarrow 2.44 \text{ mV}$

Power Analyzer Status	Successful - Wed Feb 12 15:26:26 2025
Quartus Prime Version	20.1.1 Build 720 11/11/2020 SJ Lite Edition
Revision Name	t1c_riscv_cpu
Top-level Entity Name	task5
Family	Cyclone IV E
Device	EP4CE22F17C6
Power Models	Final
Total Thermal Power Dissipation	101.77 mW
Core Dynamic Thermal Power Dissipation	2.44 mW
Core Static Thermal Power Dissipation	78.64 mW
I/O Thermal Power Dissipation	20.70 mW
Power Estimation Confidence	Low: user provided insufficient toggle rate data

Pipelined $\rightarrow$

CPU $\rightarrow 3.4 - 6.8 \text{ mV}$

Settings	Utilization	Name	Clocks (W)	Signals (W)	Data (W)	Clock Enable (W)	Set/Reset (W)	Logic (W)	BRAM (W)	I/O (W)
Summary (0.643 W, Margin: N/A)	Summary (0.643 W (10% of total))	pipelined_riscv_cpu_wrapper	<0.001	0.017	0.016	<0.001	<0.001	0.017	<0.001	<0.001
Power Supply	> 0.034 W (5% of total)	pmu (pl_riscv_cpu)	<0.001	0.005	0.005	<0.001	<0.001	0.008	<0.001	<0.001
Utilization Details	> 0.002 W (1% of total)	ila (ila_0)	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	<0.001	0.002
Hierarchical (0.051 W)	> 0.002 W (1% of total)	bufs_c0125 (BUPD05)	<0.001	0.001	0.001	<0.001	<0.001	0.001	<0.001	<0.001
Clocks (<0.001 W)	> 0.001 W (1% of total)	leaf (leaf_0)	<0.001	0.001	0.001	<0.001	<0.001	0.001	<0.001	<0.001
Signals (0.023 W)		Leaf Cells (23)								
Data (0.023 W)										
Clock Enable (<0.001 W)										
Set/Reset (<0.001 W)										
Logic (0.026 W)										
BRAM (<0.001 W)										
I/O (0.002 W)										

$$Power = CV^2f \text{ (dynamic)}$$

1 MHz $\rightarrow$ 1 Million Instruction

CPU CLOCK $\rightarrow \sim 25 \text{ MHz}$

$$T = \frac{1}{25 \times 10^6} = 40 \text{ ns} = 25 \text{ Million Instructions}$$

Energy from RF $\rightarrow$

900 MHz - 3.5 GHz

$$\lambda = \frac{c}{f}$$

$$A_{bb} = \frac{\lambda^2}{4\pi}$$

Lower bound $\rightarrow \frac{3 \times 10^8}{900 \times 10^6} = 0.33 \text{ m} \Rightarrow A_{bb} = \frac{(0.33)^2}{4\pi} = 0.0087 \text{ m}^2$

Upper bound $\Rightarrow \frac{3 \times 10^8}{3.5 \times 10^9} = 0.085 \text{ m} \Rightarrow A_{bb} = \frac{0.085^2}{4\pi} \approx 0.0006 \text{ m}^2$

Power for 900 MHz

$$P_{received} = P_{density} \times A_{bb}$$

$$P = 0.1 \text{ mW/m}^2 \times 0.0087 \approx 0.87 \text{ W}$$

Power for 3.5 GHz

$$P = 0.1 \text{ mW/m}^2 \times 0.0006 \approx 0.06 \text{ W}$$

$$\text{Total harvested power} = 0.87 + 0.06 \approx 0.93 \text{ W}$$

Storage $\rightarrow E = \frac{1}{2} CV^2$

(We want 6 mV for 1s so that we can do 1 Million instructions)

$$E = P \times t$$

$$E = 6 \text{ mV} \times 1 \text{ s} = 6 \text{ mJ}$$

$$C = \frac{2E}{V^2} = \frac{2 \times 6 \times 10^{-3}}{3^2} = 1330 \text{ nF}$$

$$\text{Time} = \frac{E}{P} = \frac{6 \times 10^{-3}}{0.93 \times 10^{-9}} \approx 6451 \text{ s} = 1.8 \text{ hrs}$$

To get 6 mV for 1 s a 1330 nF cap can be charged for 1.8 h with RF harvesting ✓ (Pipeline)
